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Computational Neuroscience

Midterm Project

This is in three parts, I wanted to stay a good distance away from a formal paper so this is a general spiel of what I might say if:

1. A first grader asked what I'm reading because that's just what first graders do when they see you holding a book.
2. A William & Mary student (perhaps in the sciences) asks what I'm reading because because they're actually interested, and
3. An adult, not in the sciences, asks what I'm reading because they feel like they should ask because I have a book in my hand.

1.

Do other animals think like us? Can other animals think like us? And if they think like us does that mean they have to be like us? What do brains do? Brains let us think, and we think so that we can solve problems and do things like eat and play games. Some animals like ants will find food and try to not get stepped on, but they don't really think like we do. Does this make us different from some animals? Ants don't have feelings like we do but they live in big colonies sort of like how we live in cities, so why do people have feelings? There is something different about humans than some animals with smaller brains and it is that we are able to know what it is like to feel something.

We know monkeys are smart because they act like us. But monkeys look like us so it might not be surprising that they are smart like us. An octopus is very different than us, on the outside and the inside, but it does a lot of things that people do too. Some people have seen an octopus play with rocks and cans. People have also seen them have feelings, like when they like one person but not the other, they show it in their actions. But just like what we said earlier, feelings aren't needed for the octopus to survive, so why does it have them?

The body of an octopus is very different than ours, especially its brain. People have their brain in their head but an octopus has its brain spread out over its entire body. It has some of its brain in all eight of its tentacles. An octopus can also see with its skin. If an octopus wants to change color to blend in with the rock behind it, it doesn't need to see the rock with its eyes, it can tell its skin to look at the rock and then the skin will change into that color. So if we had our brains in our arms and legs and could see with our skin do you wonder if we would think differently? Probably. Some people say that even though an octopus has a very different way of living than people do, it thinks similar to what a person thinks.

So if an octopus thinks like we do, but is very different than we are, that means that whatever we have in common with an octopus is very special. Having feelings and understanding the world around us is a very important part of what makes us humans and what makes them octopi. And even though our bodies and brains are very different, we can understand each other. If you see an ant it will run away from you because it can't understand you but if you ever see an octopus out in the ocean, try to reach out and touch it, because it might reach back and try to understand you, just like you.

2.

Consciousness is extremely difficult to define. There are many different approaches to try to explain what makes us 'us.' We can think about consciousness as being aware of the world around us but to some fuzzy degree. Ants are aware of what is around them; when you try to pick them up they turn around and run away but we don't usually think of a single ant as conscious. Perhaps we need to change our definition to include being aware of not only the world around us but also being aware of ourselves. That would then rule out more "simple" organisms that respond to the world but only really out of a need to survive. I believe we can agree that we, as humans, are conscious beings, and if we retroactively restricted our definition of consciousness to *only* include humans then it doesn't become an interesting problem anymore. What we should do is find a more philosophical approach to the biological evidence we have that an organism such as an octopus exhibits certain complex and curious behavior and grow our understanding of consciousness beyond our model that uses the human brain or even the human mind as a basepoint.

Let's lay down some other terms and statements that will eventually lead us to a definition of "consciousness" that, won't necessarily be agreed upon as the true meaning of consciousness, but leaves us enough wiggle room that we might include some non-humans, maybe, we'll see. As we briefly mentioned above, (almost?) all life has some awareness of the world it lives in. Insects hunt and avoid danger, even bacteria can swim toward food and away from toxins, humans can do that and a lot more. Billions of years ago when the only life on earth was very simple, perhaps no notion of consciousness existed yet, so organisms exhibited behavior that pushed them toward surviving long enough to reproduce. This tells us that these organisms *sensed* the world around them, in order to enact these behaviors. There are two subtle notions of "self" at this point in time. An organism can **sense** the world around it and can then **move** accordingly, or, an organism can **move** and then **sense** new things according to its movements. These sound like essentially the same thing but if we consider **why** an organism might move then we can see deeper into how higher cognition might arise. If some prehistoric worm senses food and then moves toward it and then its sensation of food increases, it can determine that it is heading toward food. If its sensation of food decreases after moving a bit then it can determine that it is heading away from food and thus might change directions. This is the very mechanics of bacteria that drive their motion. We agree that a single celled bacteria doesn't have any awareness of itself or the future; this is simply a system in which it is wired for survival. Now imagine a somewhat more complex organism that can move quite freely. It can decide to move, for whatever reason, and sense new things. If it can create connections between the movement of its body in space and what new senses it experiences, then these connections are actually representations of itself interacting with the world. Let's think about this level of what Godfrey-Smith calls "subjective experience."

The author of this book says that subjective experience is the idea that "life feels like something." It's hard to pin down exactly what that something is, but we can use its vagueness to point out that consciousness is a phenomenon within the realm of subjective experience. This allows us to make that distinction between insects and fish that do exhibit complex behavior and, as we can observe, have some notion of self-preservation, and humans which we argue are (more) "conscious." Insects, fish, lobsters, etc. avoid dangerous situations that jeopardize their survival, but does being damaged

actually hurt them? Do they have a notion of pain or just have the connection that if they die they can't reproduce and thus have evolved to avoid death. Here, we will say that "consciousness" in some context, is having some perception of events relating to a self that is independent of the events as they happen. This is general enough to include aspects of what we think of as relating to consciousness such as emotion, motivation, theory of mind, time/space, object permanence, and many others. Now, we can look for other organisms that share this definition with us.

To an almost spooky degree, octopi are very smart. There have been countless stories of them acting in ways that are so ... human(?) we can't help but see some connection between our minds and theirs. Minds, not brains. Humans and octopi have a common ancestor so far back in history that it's hard to believe either of us evolved from it. A tiny worm-like organism from more than 600 million years ago is the most recent organism we have in common. At a certain point in the history of evolution, the worm would evolve in one way to end up at vertebrates like dogs and fish and humans, and in another way that would end up at lobsters and snails and octopi. Because we have had 600 million years to diverge in how our brains and bodies are formed, it makes a lot of sense that we are physically very different from an octopus. Yet, how is it that they act in ways sometimes so similar to us we label it as "human-ish"? Various studies have shown that the nervous system of an octopus is extremely different from ours. The brain in their head accounts for less than half the neurons in their body, a majority of their neural processing happens in their tentacles. This leads to a variety of strange behaviors that we as humans can't even wrap our heads around.

Godfrey-Smith outlines in his book some experiments and stories explaining the strange relationship between an octopus and its tentacles. An octopus has some general executive control over its limbs, but most of the time when the octopus doesn't need all eight of them, they just do their own thing, tactilely exploring the world they're in, storing individual memories and responding to stimuli at perhaps the level of subjective experience as mentioned above, but not complete consciousness. An octopus seems to almost shepherd its tentacles as all of them are collectively used to aid in the goals of the octopus but also remain strangely independent. This is really weird and cool and we'll never be able to imagine living a life like that. In the author's words, octopi are the closest thing we have right now to meeting an intelligent alien. We have arrived at comparable levels of cognition, yet the machinery that allows it have evolved completely differently.

What does this strange relationship say about two brains being wildly different but the minds inside of them being relatively similar? What can we say about consciousness now? Again, it's a hard problem to prove whether something is conscious, not only for the reason that the definition of consciousness is quite fuzzy, but whatever definition we do put in place becomes very hard to test with organisms we can't talk to, which is all of them. We have anecdotal evidence of humans interacting with octopi that claim to see undeniable evidence of connection between two conscious minds. We also have biological and psychological evidence that these cephalopods have extremely complex and capable nervous systems. Is this enough to say that an octopus is conscious? It's difficult to know. But if we look for consciousness outside the framework that we as humans have experienced it, we might not have to go to outerspace to find it.

3.

Are we the only conscious beings on earth? We could argue that chimps are conscious, they are so close to us that it would be hard to observe their behavior that is so similar to ours and not conclude that they think in comparable ways to us humans. What if we cut out mammals? Or even vertebrates in general? Would there be anything else left on the list of organisms that have comparable cognition to a human? Some would say yes. Cephalopods, and above all of them the octopus, exhibit extremely complex behaviors that when seen in person, some people cannot help but see a connection between us and them. When we interact with many animals, it seems that their interaction with us is purely based on their instinct of survival. Birds fly away when we get too close to them, but ducks will approach us when we give them bread. It would be an interesting experience if we interacted with some animal that was so different from us at a physical level but had a mind similar to ours, if it interacted with us in a way that was not only evaluating us as a predator or as prey. Stories like this about octopi have been around ever since people started studying them. In addition to being able to solve complex problems like opening jars to find food or using tools (just as some birds have recently been shown to be capable of), there have been many accounts from researchers of an octopus seeming to play with something that clearly is not food just for the sheer entertainment, or an octopus squirting water at lights outside their tank to break them because the glare was obnoxious. Instances like these and countless others lead us to believe that an octopus has some sense that it is an octopus, understands the world around it, and that there are beings occupying the same world as them. This is particularly fascinating because of how different their brains are from ours.

The common ancestor of humans and octopi existed more than 600 million years ago. Because the fork of how we developed as organisms happened so long ago, we have had much more time to evolve in parallel to each other and thus build wildly different nervous systems. This is of particular interest because it makes sense that humans and chimps have similar cognition because we are very close relatives evolutionarily speaking, but we have no reason to believe that something with such a different journey than us could be so similar in how it thinks. This raises the question of trying to distinguish the brain versus the mind. If we had no evidence that beings with different brains than ours could not be conscious like we are, then we would rule out that any intelligent life could exist. So, we are trying to extend our definition of consciousness to something that is not dependent on the physical “brain” and come to some better understanding of how “other minds” work.

We could never imagine what it is like to be an octopus. In addition to having a vastly different brain than us, it's not even all inside their heads. A majority of an octopus' neurons are in its tentacles. Because of this their appendages are far more independent than how we perceive ours to be. Octopus tentacles have a lot more than just the mechanical facilitation of grabbing things to eat and mating and fighting, they can taste and see as well. If an octopus reaches out to touch your hand, it is tasting with the receptors on its skin to get a better idea of what you are. And the skin of an octopus cannot actually “see” in the sense that its eyes do; they do not form images but they do respond to light which sends signals to the tentacle to act a certain way. For

an octopus to change color, it does not actually need to see with its eyes what the color and texture of the surface is like, its skin responds to the lights that it is close to and then, if the octopus chooses, allows the skin to change color in accordance with what it senses. This individual experience and capabilities of the octopus tentacles leads to strange phenomena where different tentacles of an octopus can store memories and act somewhat independently of the being their attached to, but then get overridden with whatever executive control the octopus imposes on them, like to reach out and grab food. It is impossible to imagine experiencing the world the way an octopus does because *how* we imagine anything is based on how we experience the world currently. Just like we cannot think of a new color, we definitely cannot think of a new way of thinking. So we should not think of consciousness as something unique to how we are constructed, we should look at all the different forms consciousness might come in and by doing this we can approach the idea of consciousness from a more meaningful perspective.